Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Create a physical network topology in Cisco Packet Tracer.

Task: Document the steps for configuring a small network.

❖ **1. Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.**

➢ **Aim**

- To create a small home network topology in Cisco Packet Tracer consisting of three PCs and one wireless router, configure the devices, and verify network connectivity.

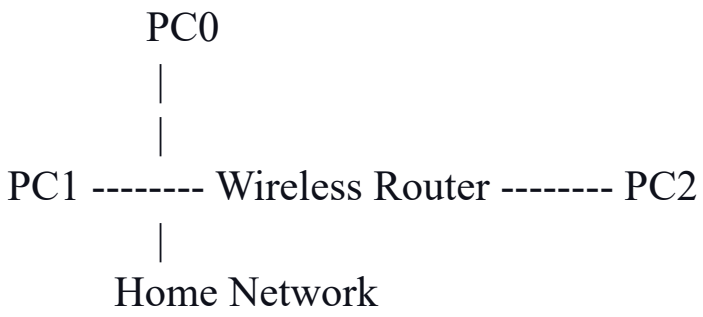
➢ **Introduction**

- A **network topology** is the physical or logical arrangement of devices in a computer network. It describes how computers, routers, switches, and other devices are connected and communicate with one another.
- Cisco Packet Tracer is a network simulation tool used to design, configure, and test computer networks in a virtual environment.
- In this practical, a simple home network is created using three PCs and one wireless router. The PCs are connected to the router through Ethernet

cables, and the router provides network connectivity and can provide wireless access to compatible devices.

- **Tools and Requirements**
  - Cisco Packet Tracer software
  - Three PC devices
  - One wireless router
  - Ethernet cables
  - A computer or laptop

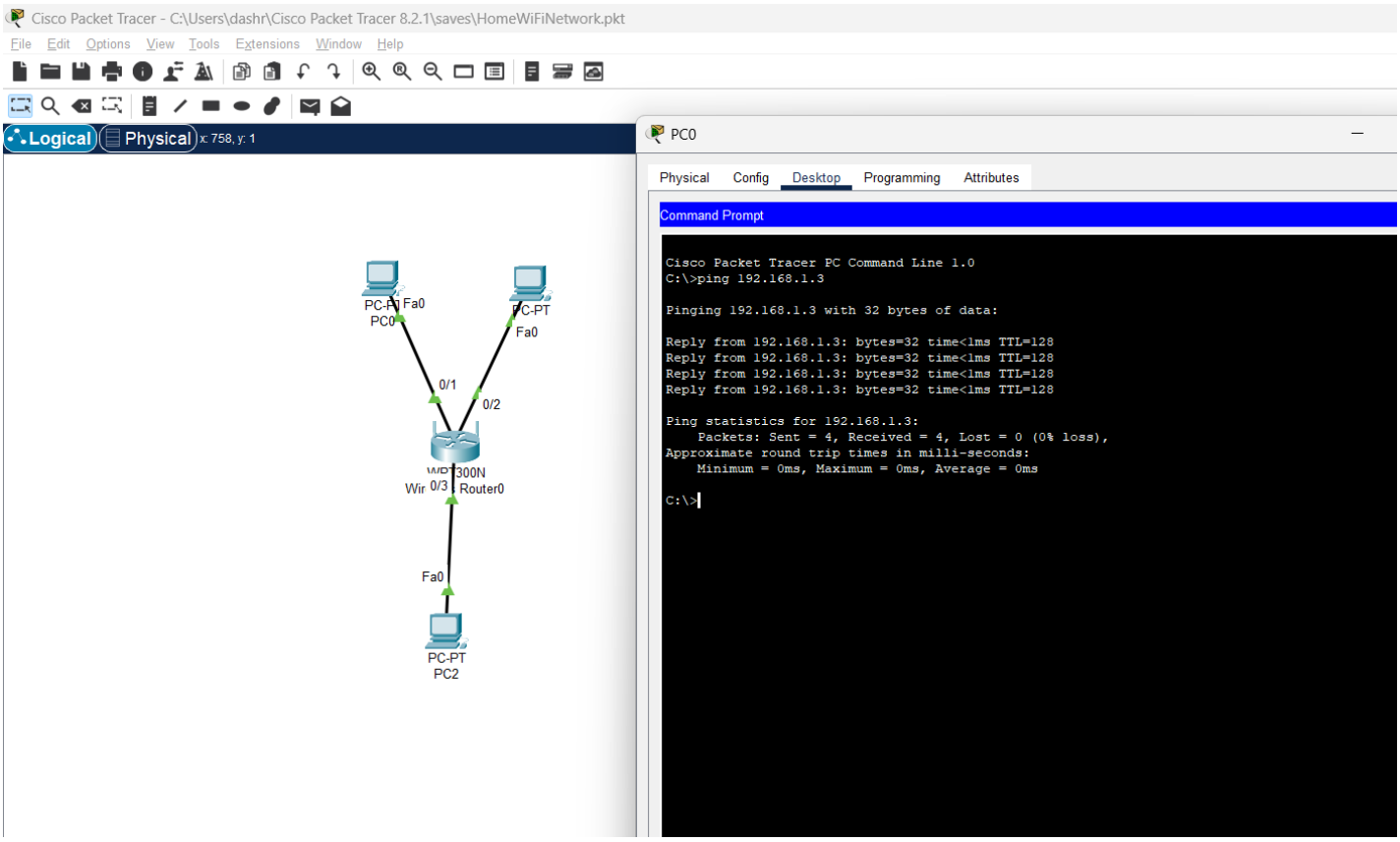
- **Network Topology**
- The network consists of one wireless router connected to three PCs.
- **Topology Diagram:**



- The diagram represents the logical arrangement of the devices. In Cisco Packet Tracer, each PC is connected to a separate LAN Ethernet port on the wireless router using a copper straight-through cable.

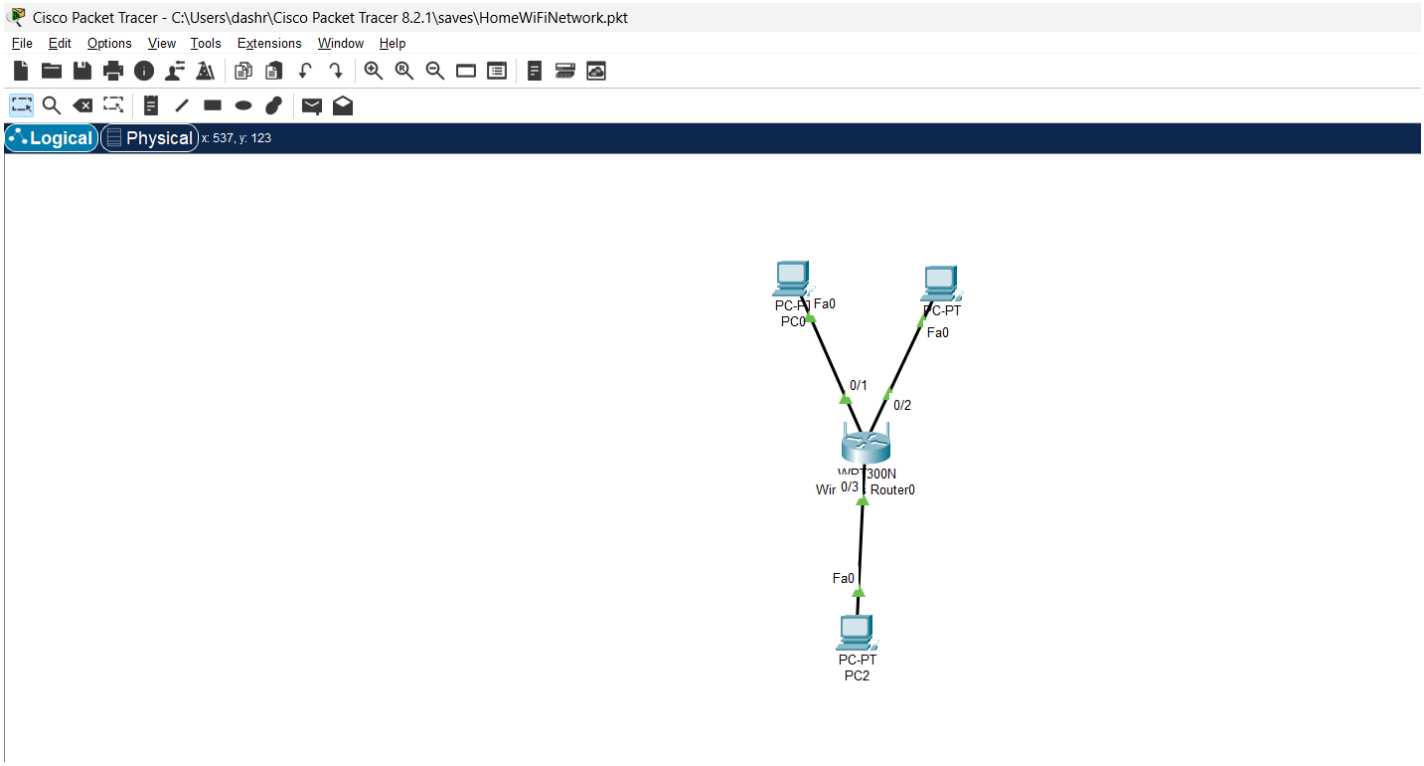
- **Procedure**
- **Step 1: Create a New Project**
  1. Open Cisco Packet Tracer.
  2. Create a new project.
  3. Add the required network devices to the workspace.
  4. Save the project as HomeWiFiNetwork.pkt.
- **Step 2: Add Network Devices**
  1. Add three PCs to the workspace.
  2. Add one wireless router.
  3. Arrange the devices neatly so that their connections are easy to understand.
- **Step 3: Connect the Devices**
  1. Select the **Connections** tool.
  2. Choose a **Copper Straight-Through** cable.
  3. Connect PC0 to a LAN port on the wireless router.
  4. Connect PC1 to another LAN port.
  5. Connect PC2 to a third LAN port.
- The router's Internet/WAN port should not be used for these three PC-to-LAN connections.
- **Step 4: Configure IP Addresses**
  1. Open each PC.
  2. Select **Desktop → IP Configuration**.
  3. Select **DHCP** if the wireless router's DHCP server is enabled.
  4. Check that each PC receives a valid IP address, subnet mask, and default gateway.
- If DHCP is unavailable, configure suitable unique IP addresses manually within the router's LAN subnet.

- **Step 5: Verify Network Connectivity**
  1. Open PC0.
  2. Select **Desktop → Command Prompt**.
  3. Use the ipconfig command to check its IP configuration.
  4. Use the ping command to test connectivity to another PC's IP address.
  5. Repeat the test from another PC if required.
- For successful communication, the PCs must have compatible IP addresses and subnet masks, and the router must provide the appropriate local network connectivity.



➢ Step 6: Save the Project

1. Click **File → Save As**.
2. Enter the file name HomeWiFiNetwork.pkt.
3. Save the project in the required folder.
4. Reopen the saved project if necessary to confirm that the topology is preserved.



➢ Device Configuration Summary

Device	Quantity	Connection
Wireless Router	1	Central network device

PC0	1	Ethernet connection to router LAN
PC1	1	Ethernet connection to router LAN
PC2	1	Ethernet connection to router LAN
Copper Straight-Through Cable	3	Connects each PC to the router

- **Result**
- A small home network topology consisting of three PCs and one wireless router was created in Cisco Packet Tracer. The project was saved as HomeWiFiNetwork.pkt, and the IP configuration and connectivity can be verified using the available simulation tools.

❖ 2. Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

➢ Aim

- To assign static IP addresses to each PC in a Cisco Packet Tracer network, configure the same subnet mask, and set the default gateway to the router's IP address.

➢ Introduction

- An **IP address** is a logical address used to identify a device on a network. Every device on the same network must have a unique IP address to communicate with other devices.
- A **static IP address** is an IP address manually assigned to a device. A **subnet mask** identifies the network portion of an IP address, while a **default gateway** allows devices to communicate with networks outside their local subnet.
- In this practical, three PCs are connected to a WRT300N wireless router in Cisco Packet Tracer. Static IP addresses are assigned to all three PCs so that they belong to the same subnet.

➢ Network Configuration Details

- **Network Address:** 192.168.10.0/24
- **Subnet Mask:** 255.255.255.0
- **Router IP Address (Default Gateway):** 192.168.10.1
- **Number of PCs:** 3

- **Static IP Address Assignment Table**

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.10.1	255.255.255.0	Not applicable
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1

- **Procedure**
- **Step 1: Open Cisco Packet Tracer**
- Open the existing Packet Tracer project containing three PCs connected to a WRT300N wireless router.
- **Step 2: Configure the Router**
  1. Click the wireless router.
  2. Open the **GUI** tab, if available.
  3. Open the router's LAN or local network setup section.
  4. Set the router's local IP address to 192.168.10.1.
  5. Set the subnet mask to 255.255.255.0.
  6. Apply or save the settings.
- **Step 3: Configure PC0**
  1. Click PC0.
  2. Open **Desktop → IP Configuration**.

3. Select **Static**.
4. Enter the following values:
   - IP Address: 192.168.10.2
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.10.1

➢ **Step 4: Configure PC1**

➢ Repeat the same steps for PC1 and enter:
- IP Address: 192.168.10.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

➢ **Step 5: Configure PC2**

➢ Repeat the same steps for PC2 and enter:
- IP Address: 192.168.10.4
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

➢ **Step 6: Save the Project**

➢ Click **File → Save** to save the updated Packet Tracer project.

➢ **6. Verification**

➢ To verify the configuration:

1. Click PC0 and open **Desktop → Command Prompt**.
2. Enter the following command:
   ipconfig
3. Check that PC0 displays the correct IP address, subnet mask, and default gateway.
4. Test communication with PC1 by entering:
   ping 192.168.10.3

5. Test communication with PC2 by entering:
   ping 192.168.10.4

- Successful ping replies indicate that the PCs can communicate over the local network. If the router's LAN address is configured correctly, its connectivity can also be checked using:
- **ping 192.168.10.1**
- **Result**
- Static IP addresses were assigned to all three PCs in the 192.168.10.0/24 subnet. The same subnet mask and the router's IP address as the default gateway were configured on each PC. Network connectivity was verified using the ping command.

❖ 3. Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router. Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'.

➢ Aim

- To test network connectivity between all PCs and the router in Cisco Packet Tracer using the ping command.

➢ Introduction

- The **ping command** is a network troubleshooting tool used to check whether one device can communicate with another device over a network. It sends ICMP Echo Request messages to the destination and waits for ICMP Echo Reply messages.
- In this practical, three PCs are connected to a wireless router in Cisco Packet Tracer. The Command Prompt tool is used on each PC to verify connectivity between the PCs and the router.

➢ Network Configuration

➢ The following IP addresses are used for this practical:

Device	IP Address
Router	192.168.10.1
PC0	192.168.10.2
PC1	192.168.10.3
PC2	192.168.10.4

- **Subnet Mask: 255.255.255.0**

- **Procedure**
- **Step 1: Open the Packet Tracer Project**
- Open the saved Packet Tracer project containing three PCs connected to a wireless router. Ensure that the devices are configured with valid IP addresses in the same subnet.
- **Step 2: Test Connectivity from PC0**
  1. Click PC0.
  2. Open **Desktop → Command Prompt**.
  3. Enter the following commands one by one:

  ping 192.168.10.3
  ping 192.168.10.4
  ping 192.168.10.1
- These commands test connectivity from PC0 to PC1, PC2, and the router.
- **Step 3: Test Connectivity from PC1**
  1. Click PC1.
  2. Open **Desktop → Command Prompt**.
  3. Enter:

  ping 192.168.10.2
  ping 192.168.10.4
  ping 192.168.10.1
- These commands test connectivity from PC1 to PC0, PC2, and the router.
- **Step 4: Test Connectivity from PC2**
  1. Click PC2.
  2. Open **Desktop → Command Prompt**.
  3. Enter:

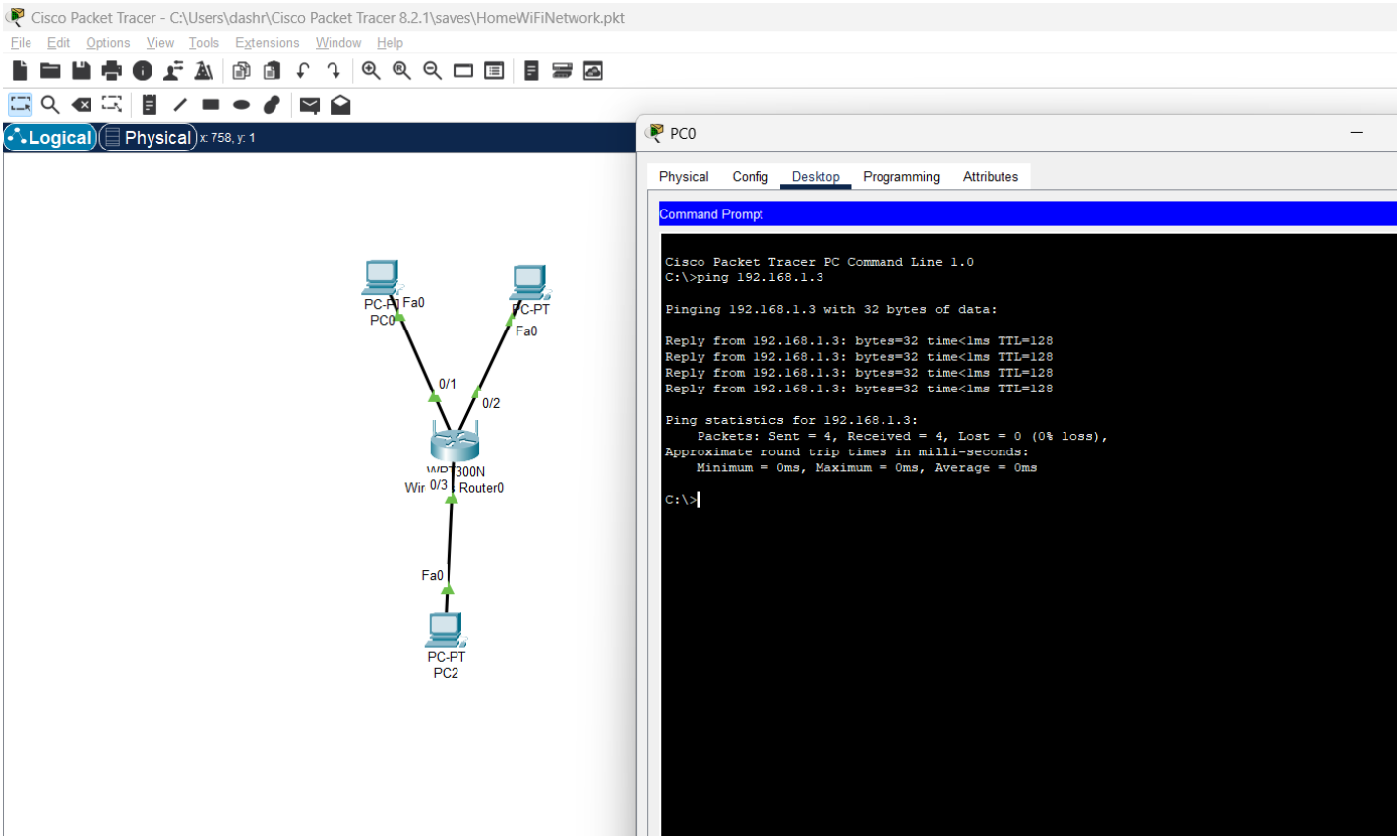
ping 192.168.10.2
ping 192.168.10.3
ping 192.168.10.1

- These commands test connectivity from PC2 to PC0, PC1, and the router.
- **Step 5: Observe the Results**
- Observe the output displayed after executing each command.
  - **Reply from:** Indicates that a response was received from the destination.
  - **Request timed out:** Indicates that no response was received within the allowed time.
  - **Destination host unreachable:** Indicates that the destination could not be reached through the available network path.
- The first ping may occasionally fail while the network resolves the destination's MAC address. If this happens, run the command again.

- **Observation Table**

Source Device	Destination Device	Expected Result
PC0	PC1	Successful ping reply
PC0	PC2	Successful ping reply
PC0	Router	Successful ping reply

PC1	PC0	Successful ping reply
PC1	PC2	Successful ping reply
PC1	Router	Successful ping reply
PC2	PC0	Successful ping reply
PC2	PC1	Successful ping reply
PC2	Router	Successful ping reply



- **Result**
- Network connectivity between the three PCs and the wireless router was tested using the ping command in Cisco Packet Tracer. Successful ping replies confirmed communication between the tested devices.

❖ **4. Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.**

➢ **Aim**

- To document the steps used to build and configure a small network in Cisco Packet Tracer, assign static IP addresses to each PC, and verify connectivity using the ping command.

➢ **Introduction**

- Cisco Packet Tracer is a network simulation tool used to create and test network topologies. In this practical, three PCs are connected to a WRT300N wireless router. Each PC is configured with a unique static IP address, a common subnet mask, and the router's LAN IP address as its default gateway. Screenshots are included as evidence of the topology and configuration.

➢ **Tools and Requirements**

- Cisco Packet Tracer
- One WRT300N wireless router
- Three PC-PT devices
- Copper Straight-Through cables
- Command Prompt and IP Configuration tools

- **Network IP Addressing Plan**

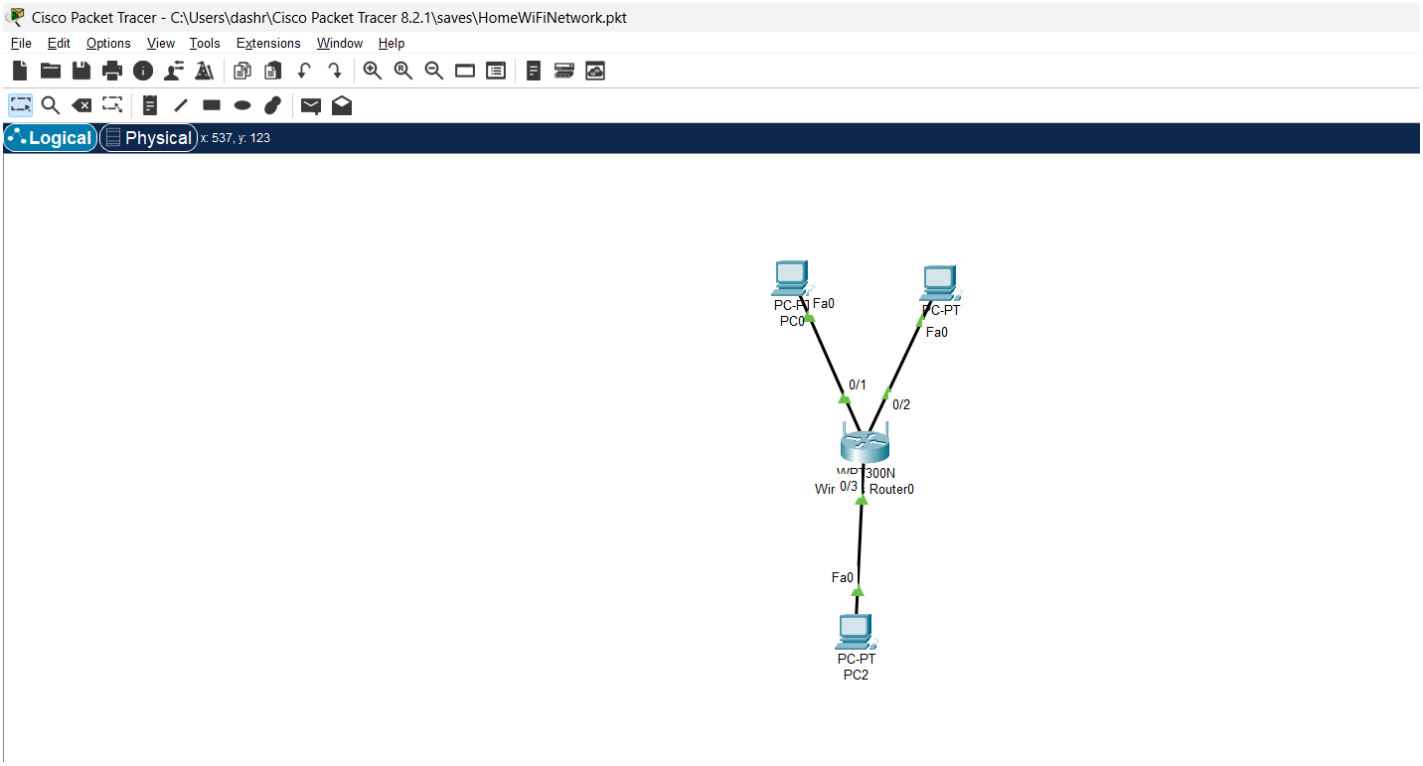
Device	IP Address	Subnet Mask	Default Gateway
Router (LAN)	192.168.0.1	255.255.255.0	Not applicable
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1

- **Procedure**
- **Step 1: Create and save the project.** Open Cisco Packet Tracer, create a new project, and save it as HomeWiFiNetwork.pkt.
- **Step 2: Add the devices.** Place three PC-PT devices and one WRT300N wireless router in the workspace. Arrange the PCs around the router so the topology is easy to read.
- **Step 3: Connect the devices.** Choose Connections and select Copper Straight-Through. Connect PC0 FastEthernet0, PC1 FastEthernet0, and PC2 FastEthernet0 to separate Ethernet LAN ports on the wireless router. Do not use the router's Internet/WAN port for these local PC connections.
- **Step 4: Configure the router LAN address.** Open the router configuration and set its LAN/local IP address to 192.168.10.1 with subnet mask 255.255.255.0. Apply the settings. The router must use this address for the gateway values below.

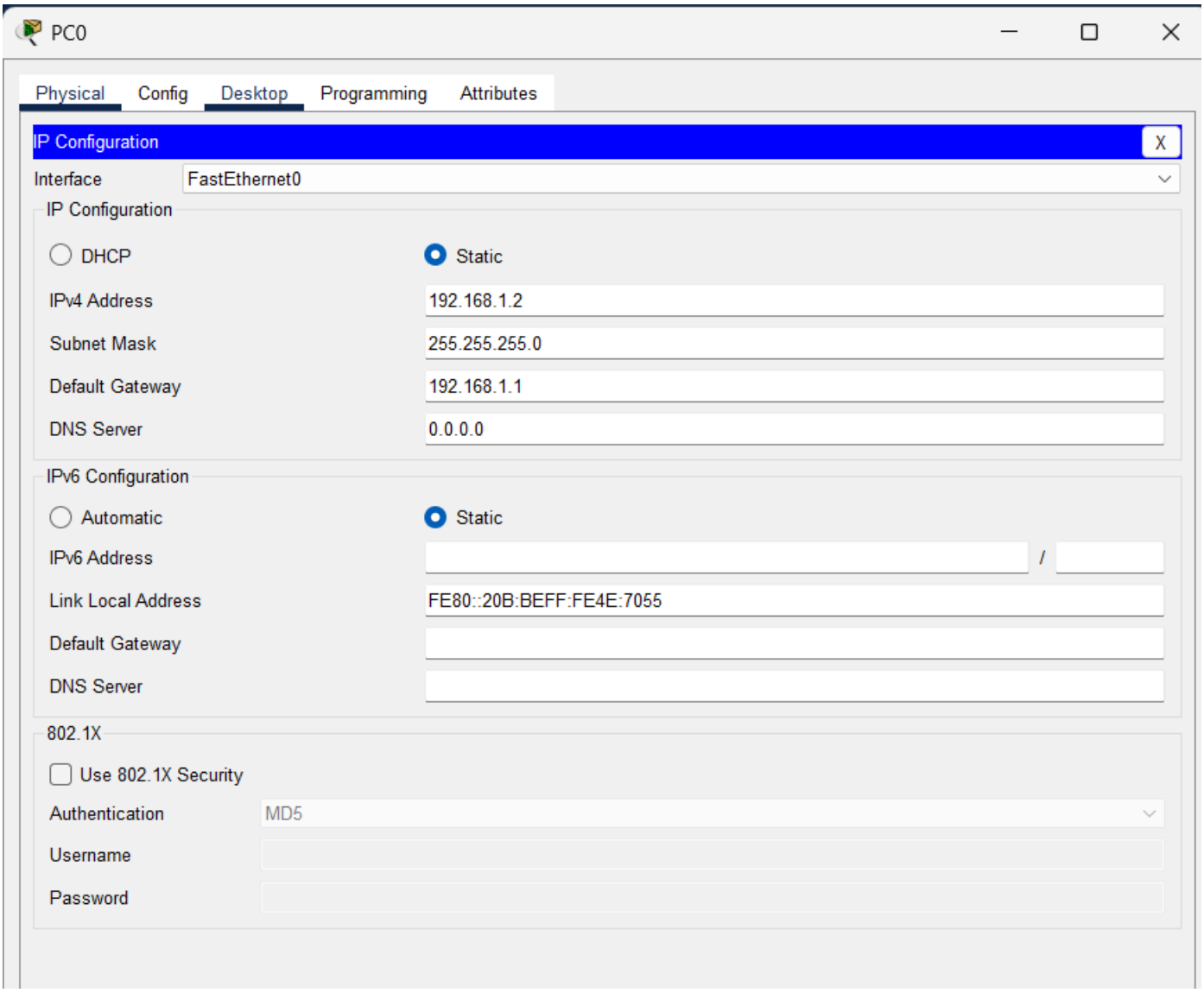
- **Step 5: Configure PC0.** Click PC0, open Desktop > IP Configuration, select Static, and enter IP address 192.168.10.2, subnet mask 255.255.255.0, and default gateway 192.168.10.1.
- **Step 6: Configure PC1.** Click PC1, open Desktop > IP Configuration, select Static, and enter IP address 192.168.10.3, subnet mask 255.255.255.0, and default gateway 192.168.10.1.
- **Step 7: Configure PC2.** Click PC2, open Desktop > IP Configuration, select Static, and enter IP address 192.168.10.4, subnet mask 255.255.255.0, and default gateway 192.168.10.1.
- **Step 8: Test connectivity.** Open Desktop > Command Prompt on a PC and ping the other PCs and the router. For example, from PC0 run ping 192.168.10.3, ping 192.168.10.4, and ping 192.168.10.1. Record the actual results.
- **Step 9: Save the project.** Save the updated Packet Tracer file using File > Save.

- **Screenshots**
- Insert screenshots captured from your own Packet Tracer project in the spaces below. Make sure the displayed IP addresses match the addressing table and the actual project. Do not use sample or illustrative images as evidence of your own configuration.

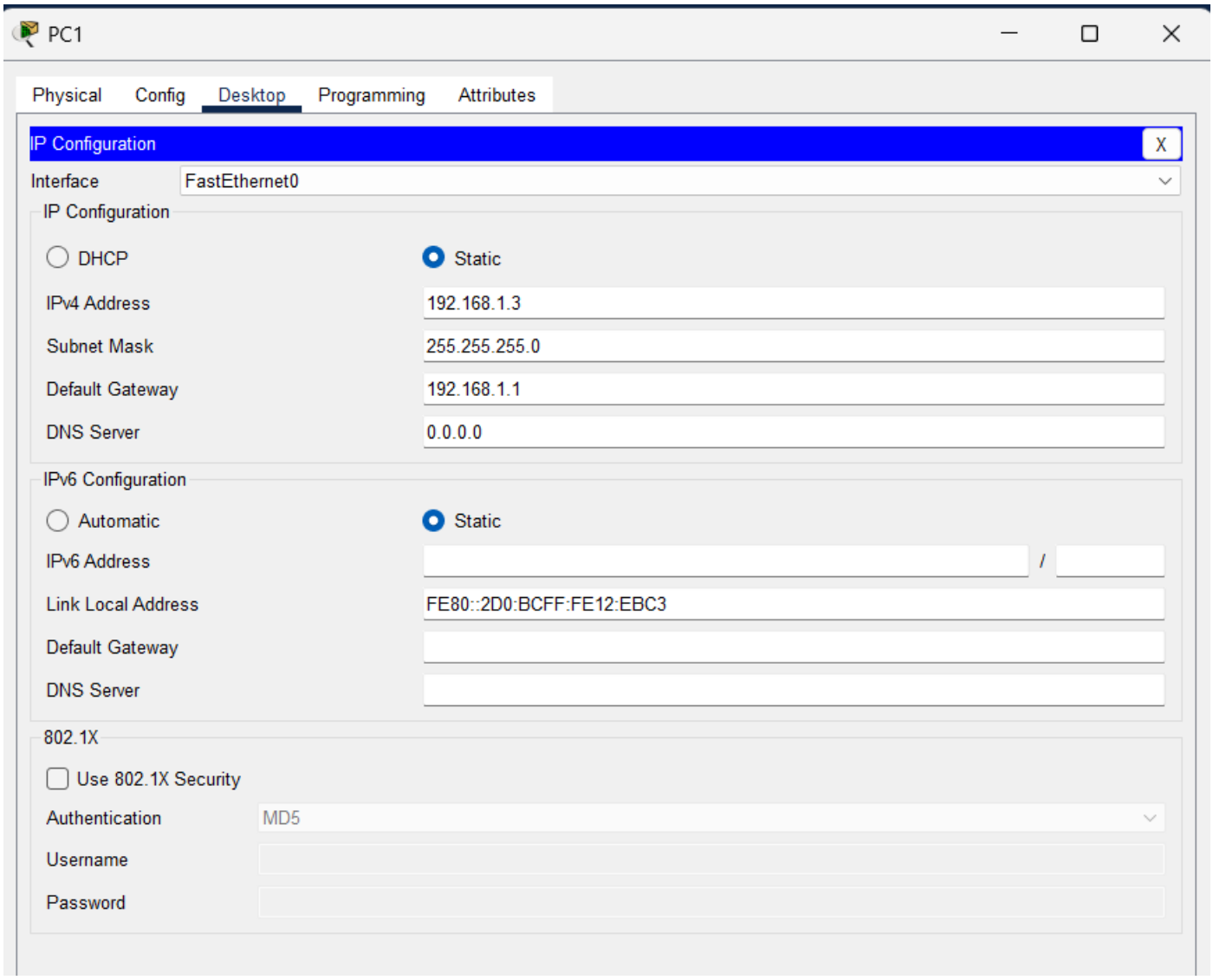
- **Screenshot 1: Completed Network Topology**



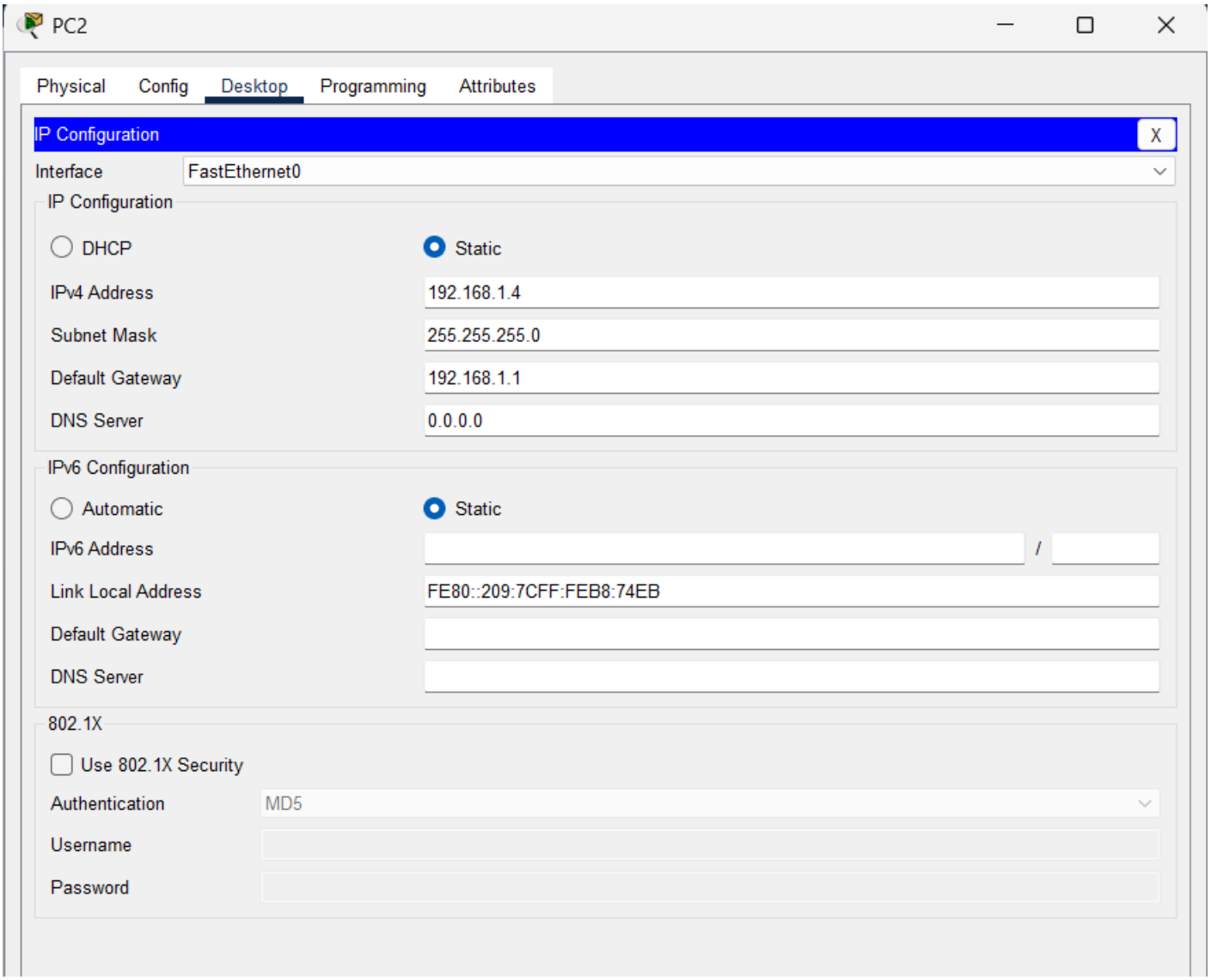
- Capture the Packet Tracer workspace showing PC0, PC1, PC2, the WRT300N router, and all three connections. Insert the screenshot here.
- **Screenshot 2: PC0 IP Configuration**



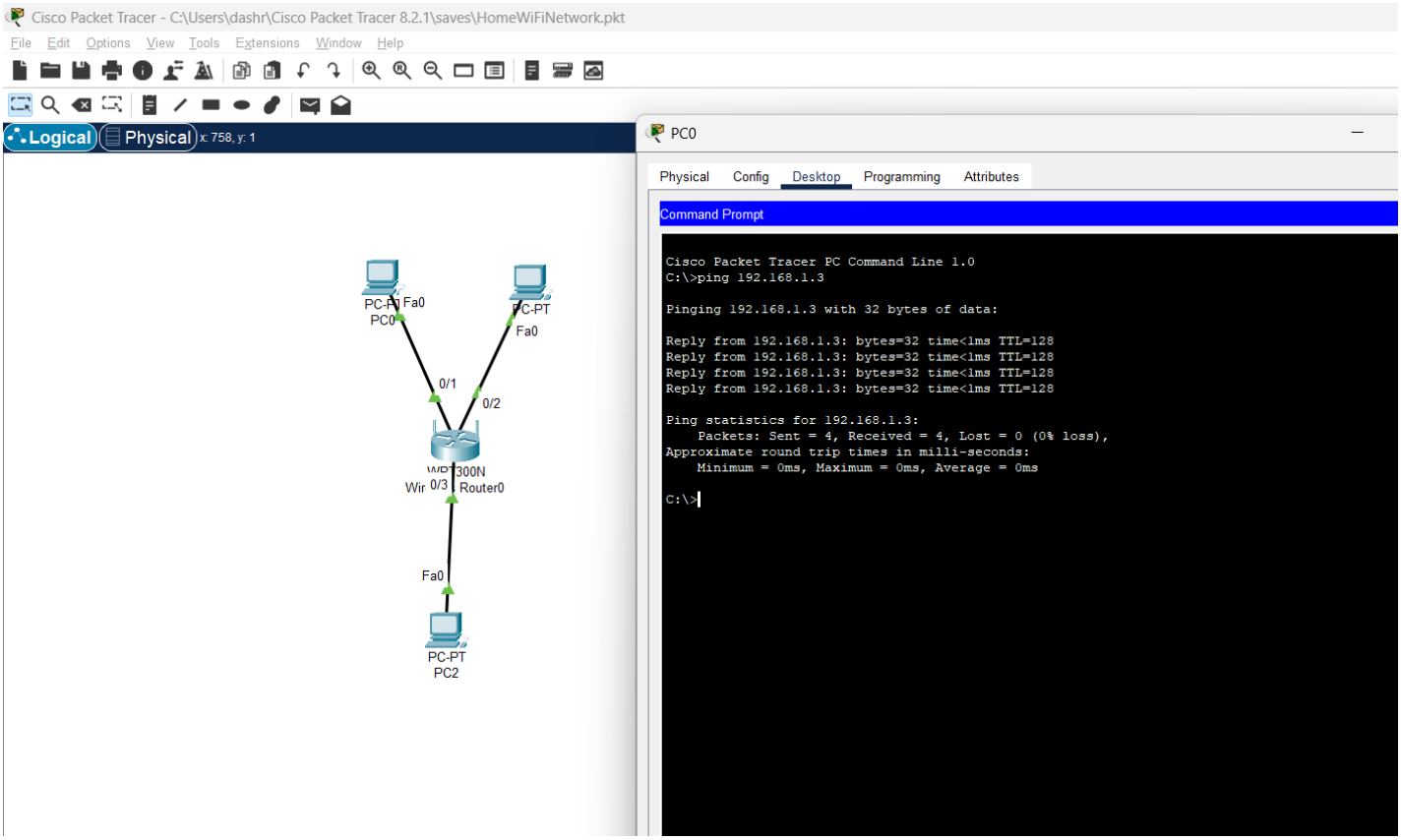
- Open PC0 > Desktop > IP Configuration and capture the IP address, subnet mask, and default gateway. Insert the screenshot here.
- **Screenshot 3: PC1 IP Configuration**



- Open PC1 > Desktop > IP Configuration and capture the IP address, subnet mask, and default gateway. Insert the screenshot here.
- **Screenshot 4: PC2 IP Configuration**



- Open PC2 > Desktop > IP Configuration and capture the IP address, subnet mask, and default gateway. Insert the screenshot here.
- **Screenshot 5: Connectivity Test (Optional but Recommended)**



- Capture successful ping replies from the Command Prompt. Insert the screenshot here.

- **Verification**
  - Each PC has a unique IP address.
  - All PCs use the subnet mask 255.255.255.0.
  - Each PC uses the router LAN address as its default gateway.
  - The PCs are connected to the router's LAN ports.

- Ping results are checked and recorded from the actual Packet Tracer project.

➢ **Result**

➢ A small network containing three PCs and a WRT300N wireless router was created in Cisco Packet Tracer. Static IP addressing was documented for each PC, and connectivity was tested using the ping command. Screenshots of the topology and each PC's IP Configuration window provide evidence of the completed practical.